

Exam 5 Review

(1) Line Integrals

- Consider the function $f(x, y) = x - y$ and the curve C which connects the points $(1, 1)$ to $(2, 3)$ linearly.

We first parametrize the curve C as $\mathbf{r}(t) = \langle t, 2t - 1 \rangle$ where $1 \leq t \leq 2$

- a) Fully setup an integral to represent $\int_C f(x, y) ds$.

$$\begin{aligned}\int_C f(x, y) ds &= \int_1^2 (t - (2t - 1)) \sqrt{(1)^2 + (2)^2} dt \\ &= \int_1^2 (1 - t) \sqrt{5} dt\end{aligned}$$

- b) Calculate $\int_C f(x, y) dx$.

$$\begin{aligned}\int_C f(x, y) dx &= \int_1^2 (t - (2t - 1)) 1 dt \\ &= \int_1^2 (1 - t) dt = \left(t - \frac{1}{2} t^2 \right) \Big|_1^2 = (2 - 2) - \left(1 - \frac{1}{2} \right) = -\frac{1}{2}\end{aligned}$$

- c) Calculate $\int_C f(x, y) dy$.

$$\begin{aligned}\int_C f(x, y) dy &= \int_1^2 (t - (2t - 1)) 2 dt \\ &= \int_1^2 2(1 - t) dt = 2 \left(t - \frac{1}{2} t^2 \right) \Big|_1^2 = 2(2 - 2) - 2 \left(1 - \frac{1}{2} \right) = -1\end{aligned}$$

- Consider the vector field $\mathbf{F}(x, y) = \left\langle xy, \frac{1}{2}x^2 \right\rangle$ and the curve C which traces outlines the region in the first quadrant determined by $y = x^2$ & $y = x$ in a counterclockwise fashion.
- a) Determine $\int_C \mathbf{F} \cdot d\mathbf{r}$ using techniques from section 5.2.

We note that C is made up of two curves C_1 & C_2 which we parameterize as $\mathbf{r}_1(t) = \langle t, t^2 \rangle$ where $0 \leq t \leq 1$ and $\mathbf{r}_2(t) = \langle 1-t, 1-t \rangle$ where $0 \leq t \leq 1$.

Therefore,

$$\begin{aligned}
 \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_0^1 \left\langle xy, \frac{1}{2}x^2 \right\rangle \cdot \langle 1, 2t \rangle dt + \int_0^1 \left\langle xy, \frac{1}{2}x^2 \right\rangle \cdot \langle -1, -1 \rangle dt \\
 &= \int_0^1 \left\langle t^3, \frac{1}{2}t^2 \right\rangle \cdot \langle 1, 2t \rangle dt + \int_0^1 \left\langle (1-t)^2, \frac{1}{2}(1-t)^2 \right\rangle \cdot \langle -1, -1 \rangle dt \\
 &= \int_0^1 (t^3 + t^3) dt + \int_0^1 \left[-(1-t)^2 - \frac{1}{2}(1-t)^2 \right] dt \\
 &= \left(\frac{1}{2}t^4 \right) \Big|_0^1 + \left(\frac{1}{2}(1-t)^3 \right) \Big|_0^1 \\
 &= \frac{1}{2} + \left(0 - \frac{1}{2} \right) \\
 &= 0
 \end{aligned}$$

- b) Determine $\int_C \mathbf{F} \cdot d\mathbf{r}$ using the Fundamental Theorem of Line Integrals.

We note that C is a piecewise smooth curve and $f(x, y) = \frac{1}{2}x^2y$ is the potential function for $\mathbf{F}$.

Therefore, by the FTOLI we have that $\int_C \mathbf{F} \cdot d\mathbf{r} = f(0, 0) - f(0, 0) = 0$.

We could have also concluded reached this conclusion because in a conservative vector field if C is a closed curve then $\int_C \mathbf{F} \cdot d\mathbf{r}$ always equals 0.

c) Determine $\int_C \mathbf{F} \cdot d\mathbf{r}$ using Green's Theorem.

We note that C is a positively oriented, piecewise smooth, simple, closed curve. We see that $\mathbf{F}$ clearly has continuous partials on the region D enclosed by C . Therefore, by Green's Theorem, we

$$\text{know that } \int_C \mathbf{F} \cdot d\mathbf{r} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \int_0^1 \int_{x^2}^x (x - x) dy dx = \int_0^1 \int_{x^2}^x 0 dy dx = 0$$

(2) Surface Integrals

- Consider the function $f(x, y, z) = xyz$ and the surface σ given by $z = x^2 + y^2$ & $z = 9$ which encloses a solid region R and is oriented outward.

a) Set up, but do not evaluate, an integral to compute $\iint_{\sigma} f(x, y, z) dS$ using a parameterization of σ .

σ is composed of two surfaces and so we will label the portion of σ that corresponds to $z = x^2 + y^2$ as σ_1 and the portion that corresponds to $z = 9$ as σ_2 .

We parameterize σ_1 as $\mathbf{r}(u, v) = \langle u, v, u^2 + v^2 \rangle$ where $-3 \leq u \leq 3$ & $-3 \leq v \leq 3$.

$$\text{Then } \mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 2u \\ 0 & 1 & 2v \end{vmatrix} = \langle -2u, 2v, 1 \rangle \rightarrow \|\mathbf{r}_u \times \mathbf{r}_v\| = \sqrt{4u^2 + 4v^2 + 1}$$

We parameterize σ_2 as $\mathbf{r}(u, v) = \langle v \cos(u), v \sin(u), 9 \rangle$ where $0 \leq u \leq 2\pi$ & $0 \leq v \leq 3$.

Then

$$\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -v \sin(u) & v \cos(u) & 0 \\ \cos(u) & \sin(u) & 0 \end{vmatrix} = \langle 0, 0, -v \sin^2(u) - v \cos^2(u) \rangle = \langle 0, 0, -v \rangle \rightarrow \|\mathbf{r}_u \times \mathbf{r}_v\| = \sqrt{0 + 0 + v^2} = v$$

Therefore, we have that

$$\begin{aligned}
 \iint_{\sigma} f(x, y, z) dS &= \iint_{\sigma_1} f(x, y, z) dS + \iint_{\sigma_2} f(x, y, z) dS \\
 &= \int_{-3}^3 \int_{-3}^3 xyz \|\mathbf{r}_u \times \mathbf{r}_v\| du dv + \int_0^3 \int_0^{2\pi} xyz \|\mathbf{r}_u \times \mathbf{r}_v\| du dv \\
 &= \int_{-3}^3 \int_{-3}^3 uv(u^2 + v^2) \sqrt{4u^2 + 4v^2 + 1} du dv + \int_0^3 \int_0^{2\pi} 9v^3 \cos(u) \sin(u) du dv
 \end{aligned}$$

- b) Set up, but do not evaluate, an integral to compute $\iint_{\sigma} f(x, y, z) dS$ without using a parameterization of σ .

σ is composed of two surfaces and so we will label the portion of σ that corresponds to $z = x^2 + y^2$ as σ_1 and the portion that corresponds to $z = 9$ as σ_2 .

We also note that both σ_1 & σ_2 both exist over the region $D = \{(r, \theta) \mid 0 \leq r \leq 3 \text{ \& } 0 \leq \theta \leq 2\pi\}$.

Therefore, we have that

$$\begin{aligned}
 \iint_{\sigma} f(x, y, z) dS &= \iint_{\sigma_1} f(x, y, z) dS + \iint_{\sigma_2} f(x, y, z) dS \\
 &= \int_0^{2\pi} \int_0^3 xyz \sqrt{(2x)^2 + (2y)^2 + 1} r dr d\theta + \int_0^{2\pi} \int_0^3 xyz \sqrt{0^2 + 0^2 + 1} r dr d\theta \\
 &= \int_0^{2\pi} \int_0^3 r^5 \cos(\theta) \sin(\theta) \sqrt{4r^2 + 1} dr d\theta + \int_0^{2\pi} \int_0^3 r^5 \cos(\theta) \sin(\theta) dr d\theta
 \end{aligned}$$

- c) Set up, but do not evaluate, an integral to compute $\iint_{\sigma} \mathbf{F} \cdot d\mathbf{S}$ by calculating the flux through each piece of the surface where $\mathbf{F} = \langle 1, x, z \rangle$.

σ is composed of two surfaces and so we will label the portion of σ that corresponds to $z = x^2 + y^2$ as σ_1 and the portion that corresponds to $z = 9$ as σ_2 .

For σ_1 we calculate $G(x, y, z) = z - (x^2 + y^2) \rightarrow \vec{\nabla}G = \langle -2x, -2y, 1 \rangle$.

Since we have downward normals on this portion of the surface, we will instead use $-\vec{\nabla}G = \langle 2x, 2y, -1 \rangle$.

For σ_2 we calculate $G(x, y, z) = z - 9 \rightarrow \vec{\nabla}G = \langle 0, 0, 1 \rangle$.

Since we have upward normals on this portion of the surface, we will use $\vec{\nabla}G = \langle 0, 0, 1 \rangle$.

We also note that both σ_1 & σ_2 both exist over the region $D = \{(r, \theta) \mid 0 \leq r \leq 3 \text{ \& } 0 \leq \theta \leq 2\pi\}$.

Therefore, we have that

$$\begin{aligned}
 \iint_{\sigma} \mathbf{F} \cdot d\mathbf{S} &= \iint_{\sigma_1} \mathbf{F} \cdot d\mathbf{S} + \iint_{\sigma_2} \mathbf{F} \cdot d\mathbf{S} \\
 &= \int_0^{2\pi} \int_0^3 \mathbf{F} \cdot -\vec{\nabla}G r dr d\theta + \int_0^{2\pi} \int_0^3 \mathbf{F} \cdot \vec{\nabla}G r dr d\theta \\
 &= \int_0^{2\pi} \int_0^3 \langle 1, x, z \rangle \langle -2x, -2y, 1 \rangle r dr d\theta + \int_0^{2\pi} \int_0^3 \langle 1, x, z \rangle \langle 0, 0, 1 \rangle r dr d\theta \\
 &= \int_0^{2\pi} \int_0^3 (-2x - 2xy + z) r dr d\theta + \int_0^{2\pi} \int_0^3 z r dr d\theta \\
 &= \int_0^{2\pi} \int_0^3 (-2r \cos(\theta) - 2r^2 \cos(\theta) \sin(\theta) + r^2) r dr d\theta + \int_0^{2\pi} \int_0^3 9r dr d\theta
 \end{aligned}$$

d) Set up, but do not evaluate, an integral to compute $\iint_{\sigma} \mathbf{F} \cdot d\mathbf{S}$ using the Divergence Theorem.

Since σ is a closed surface and $\mathbf{F}$ has continuous partials on the region E enclosed by σ , then by the Divergence Theorem,

$$\iint_{\sigma} \mathbf{F} \cdot d\mathbf{S} = \iiint_E \operatorname{div}(\mathbf{F}) dV = \iiint_E 1 dV = \int_0^{2\pi} \int_0^3 \int_{r^2}^9 1 r dz dr d\theta$$

(3) Stokes' Theorem

a) Compute $\iint_{\sigma_1} \operatorname{curl}(\mathbf{F}) \cdot d\mathbf{S}$ where σ_1 is the portion of the surface σ given by $z = x^2 + y^2$ directly. Assume the surface is oriented with upward facing normal and $\mathbf{F} = \langle 1, x, z \rangle$.

We calculate that $\operatorname{curl}(\mathbf{F}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 1 & x & z \end{vmatrix} = \langle 0, 0, 1 \rangle$

For σ_1 we calculate $G(x, y, z) = z - (x^2 + y^2) \rightarrow \vec{\nabla} G = \langle -2x, -2y, 1 \rangle$.

Since we have upward normals on the surface, we will use $\vec{\nabla} G = \langle -2x, -2y, 1 \rangle$.

Again, σ_1 exists over the region $D = \{(r, \theta) \mid 0 \leq r \leq 3 \text{ \& } 0 \leq \theta \leq 2\pi\}$.

Therefore, we have that

$$\begin{aligned} \iint_{\sigma_1} \operatorname{curl}(\mathbf{F}) \cdot d\mathbf{S} &= \int_0^{2\pi} \int_0^3 \operatorname{curl}(\mathbf{F}) \cdot \vec{\nabla} G r dr d\theta \\ &= \int_0^{2\pi} \int_0^3 \langle 0, 0, 1 \rangle \cdot \langle -2x, -2y, 1 \rangle r dr d\theta \\ &= \int_0^{2\pi} \int_0^3 r dr d\theta = \int_0^{2\pi} \frac{9}{2} d\theta \\ &= 9\pi \end{aligned}$$

- b) Compute $\iint_{\sigma_1} \text{curl}(\mathbf{F}) \cdot d\mathbf{S}$ where σ_1 is the portion of the surface σ given by $z = x^2 + y^2$ using Stokes's Theorem. Assume the surface is oriented with upward facing normal and $\mathbf{F} = \langle 1, x, z \rangle$.

We see that the boundary curve C for σ_1 is a circle of radius 3 at the height of $z = 9$.

We parameterize this curve as $\mathbf{r}(t) = \langle 3 \cos(t), 3 \sin(t), 9 \rangle$ where $0 \leq t \leq 2\pi$. Note that the orientation of σ_1 provides $\mathbf{r}(t)$ with a positive orientation. C is also a piecewise smooth, simple, closed curve and $\mathbf{F}$ has continuous partials over all of $\mathbb{R}^3$, so by Stokes' Theorem,

$$\begin{aligned}
 \iint_{\sigma_1} \text{curl}(\mathbf{F}) \cdot d\mathbf{S} &= \int_C \mathbf{F} \cdot d\mathbf{r} \\
 &= \int_0^{2\pi} \langle 1, 3 \cos(t), 9 \rangle \cdot \langle -3 \sin(t), 3 \cos(t), 0 \rangle dt \\
 &= \int_0^{2\pi} (-3 \sin(t) + 9 \cos^2(t)) dt \\
 &= \int_0^{2\pi} \left[-3 \sin(t) + \frac{9}{2} [1 + \cos(2t)] \right] dt \\
 &= 3 \cos(t) + \frac{9}{2} \left[t + \frac{1}{2} \sin(2t) \right] \Big|_0^{2\pi} \\
 &= \left(3 + \frac{9}{2} \left[2\pi + \frac{1}{2}(0) \right] \right) - \left(3 + \frac{9}{2} \left[0 + \frac{1}{2}(0) \right] \right) \\
 &= 9\pi
 \end{aligned}$$

- c) Set up, but do not evaluate, and integral to compute $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F}(x, y) = \left\langle y, \frac{1}{2}x^2, 1 \right\rangle$ and C is the boundary curve of the surface σ given by $z = 1 - x^2$ (with upward orientation) over the region $D = \{(x, y) | x^2 \leq y \leq x \text{ \& } 0 \leq x \leq 1\}$ using Stokes' Theorem.

Since $\mathbf{F}$ has continuous partials and the curve C can be positively oriented with respect to the orientation of σ , we can say that by Stokes' Theorem $\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_{\sigma} \text{curl}(\mathbf{F}) \cdot d\mathbf{S}$.

We note that $\text{curl}(\mathbf{F}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & \frac{1}{2}x^2 & 1 \end{vmatrix} = \langle 0, 0, x-1 \rangle$.

For σ we calculate $G(x, y, z) = z - (1 - x^2) \rightarrow \vec{\nabla}G = \langle 2x, 0, 1 \rangle$.

Since we have upward facing normals on this surface, we will use $\vec{\nabla}G = \langle 2x, 0, 1 \rangle$.

Therefore we have that

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \iint_{\sigma} \text{curl}(\mathbf{F}) \cdot d\mathbf{S} \\ &= \iint_D \langle 0, 0, x-1 \rangle \cdot \langle 2x, 0, 1 \rangle dA \\ &= \int_0^1 \int_{x^2}^x (x-1) dy dx \end{aligned}$$